

L1 — Equivalence Completion (Paper 8): Unitary Mixing Saturation under Admissible Redescription

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0. Scope and Non-Negotiables

This paper is eliminative. It introduces no new primitives, no new ontology, no new anchors, and no new tensor load. All results are invariant under admissible redescription.

We work with a fixed fermion generation count and fixed chiral load. No additional fermions, no additional Higgs multiplets, and no compensating sectors are permitted. All admissible freedom must survive cross-skin transport.

1. Apparent Freedom of Unitary Mixing

In broken-phase bookkeeping, fermion mixing appears encoded in arbitrary unitary matrices relating interaction eigenstates to mass eigenstates. This suggests a continuous freedom of mixing angles and phases, treated as independent physical parameters.

In gauge-phase bookkeeping, only representation content and gauge-invariant operators appear explicit, and unitary rotations are treated as basis artifacts. This creates an apparent asymmetry in where mixing data is located.

2. Admissibility Envelope

Allowed redescriptions include:

- unitary field redefinitions preserving gauge-covariant kinetic terms,
- basis changes within fixed fermion multiplets,
- reparameterizations preserving invariant operator content.

Any purported degree of freedom that cannot be transported between skins without importing new structure is inadmissible.

3. Transport of Mixing Data

Mixing matrices arise from the diagonalization of admissible Yukawa operators. They therefore encode no independent structure beyond the existence and form of those operators.

Transport from the broken-phase skin to the gauge-phase skin erases explicit mixing matrices, retaining only operator-existence conditions. Conversely, any mixing appearing in the broken phase must descend from admissible operator structure in the gauge phase.

4. Saturation of Unitary Freedom

Assume arbitrary independent unitary mixing data in the broken phase. Transport to the gauge phase.

If such data does not correspond to admissible operator structure, it cannot be represented without: (i) enlarging the fermion load, (ii) introducing compensating multiplets, or (iii) adding non-gauge-invariant terms. Each option is forbidden.

Therefore, all admissible unitary mixing freedom is exhausted by basis choice and operator diagonalization. No residual independent degrees of freedom survive transport.

5. Core Result

Theorem 1 (Unitary Mixing Saturation under Admissible Transport). *Fix the fermion generation count and chiral load. Under admissible redescription and under the global constraints (no new load, no patches), all admissible unitary mixing freedom is saturated by basis reparameterization and Yukawa operator diagonalization.*

No additional independent unitary mixing parameters exist with standing across skins.

Corollary 1 (Phase Exhaustion). *All admissible phases are either removable by redefinition or fixed by operator structure. No physical phases remain beyond those already encoded in admissible Yukawa data.*

Corollary 2 (Skin Illusion of Mixing Freedom). *The appearance of free unitary mixing parameters is a bookkeeping artifact of the broken-phase skin. Cross-skin admissibility collapses this illusion.*

6. Closure Statement

Unitary mixing does not constitute independent physical structure. It is fully exhausted by admissible operator content and basis choice under fixed load. No additional degrees of freedom survive admissible transport.